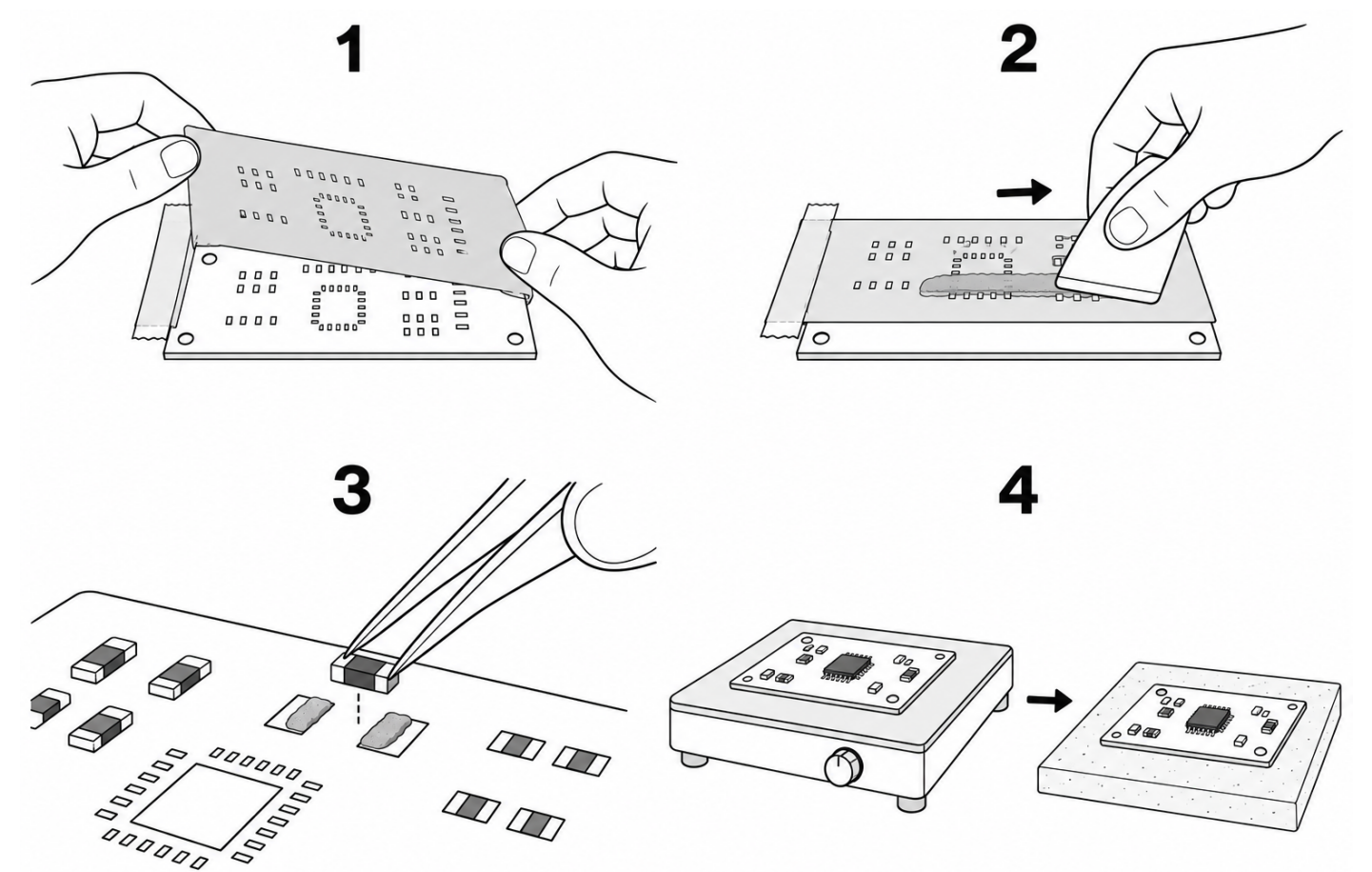


Start with the flat parts

Stencil + hot bed first. Iron after cooling. Build one sensor, one controller and one mains board.



1 ALIGN	2 PASTE	3 PLACE	4 REFLOW + COOL
Support the bare board level. Tape the 0.10 mm top stencil as a hinge; align every opening with its pad.	Pull one thin, even print. Lift vertically. Correct a smeared print before placing components.	Use labelled bags and the placement maps. Small ICs and passives first, then larger SMT parts.	Use the paste maker's temperature profile. Measure board temperature. Let solder solidify before moving.

The order for this build

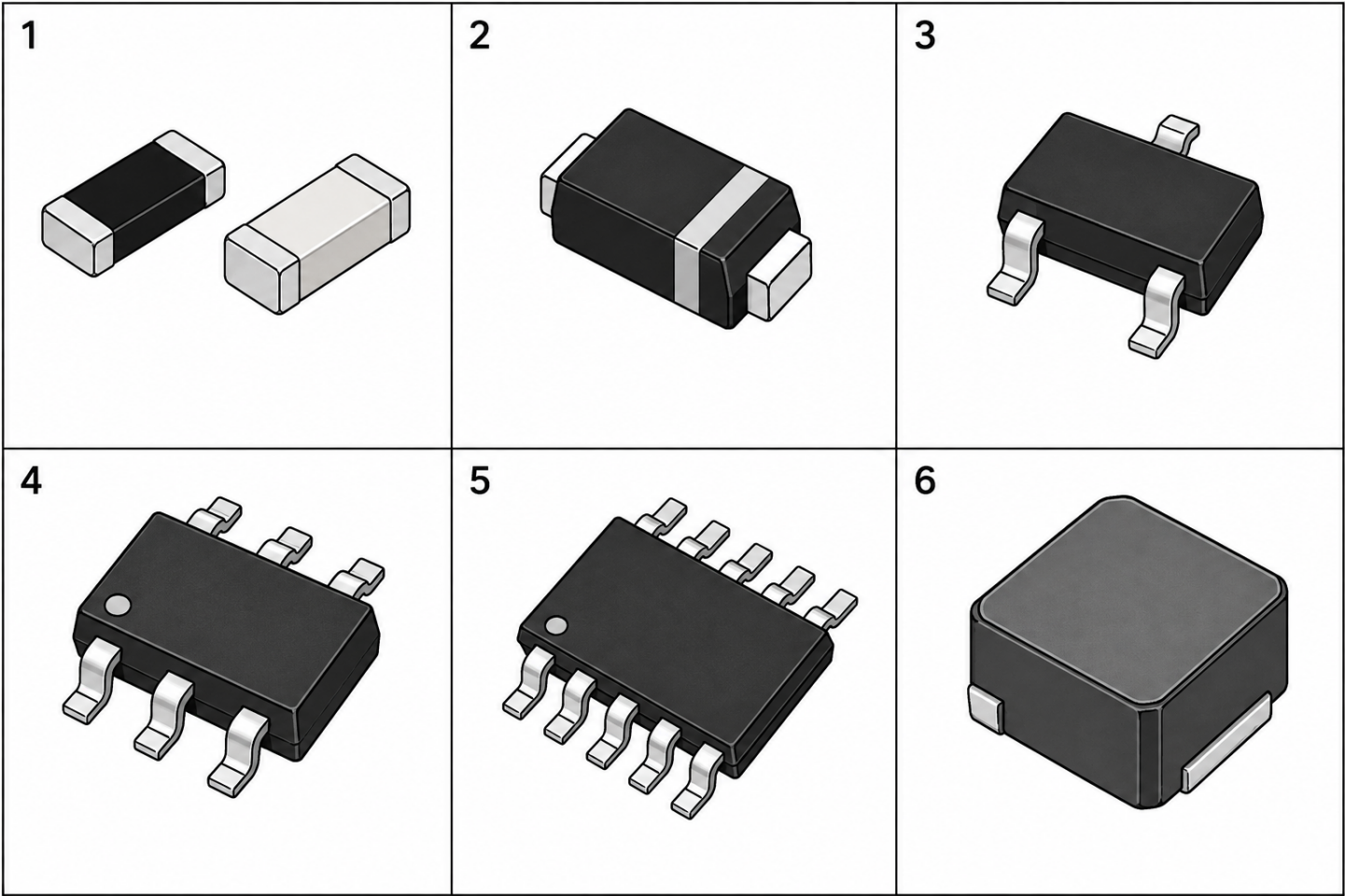
Sensor SMT → controller SMT → mains SMT → all iron work → harnesses → lower board wiring → separator → controller.
Each board is a separate hot-bed run.

Keep this beside the bench

All assembly is unpowered. Leave firmware, power-up, testing and tank mounting for our interactive sessions. Quantities here make **one set**, even though five bare boards of each type were ordered.

Recognize the SMT packages

Drawings identify package families. The bag's exact MPN determines the part; similar-looking parts are not interchangeable.



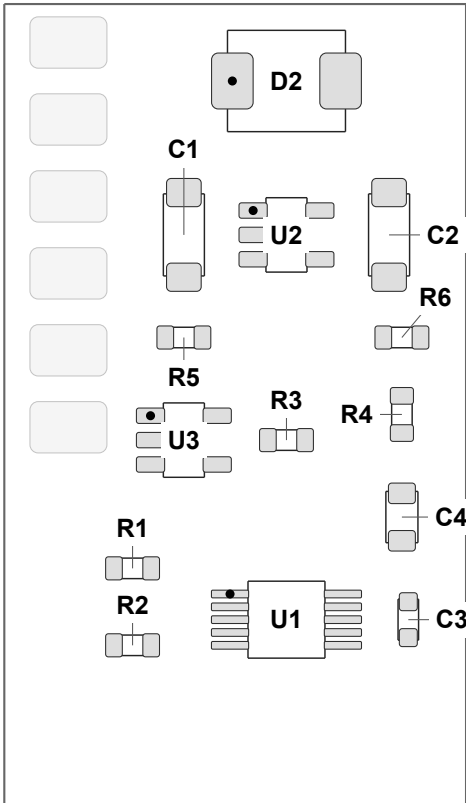
PICTURE	WHAT YOU WILL SEE	USE THE MAP FOR...
1 Chip passives	Resistors; unmarked ceramic capacitors. All are nonpolar.	Exact value and size. 0603, 0805, 1206 and 1210 are different bodies.
2 Diode	STPS2L40U and SMBJ7.0A have a cathode band.	Band at the black dot on D references.
3 Three leads	AO3400A transistors Q1–Q4 on the controller.	Two-leg side down, lone leg up in the top-view maps.
4 Small IC	Five- and six-lead ICs share this general outline.	Exact MPN and manufacturer's pin-1 identifier.
5 Fine-pitch IC	FDC1004 and FSUSB42 are ten-lead parts; TCA9517 has eight.	Pin 1. Do not use body text direction as the orientation cue.
6 Inductor	SRP5030TA-4R7M, one on each large board; nonpolar.	Controller L1 and mains L1.

Map key

All maps look down at the component side and are **never mirrored**. A black dot marks IC/MOSFET **pin 1** or diode/LED **cathode**. Pale outlines are neighbouring parts. Drawings are enlarged, not solder stencils.

Sensor / all 14 parts reflow

Electronics face toward you, RIM edge at the top. The six left-hand lands get wires later, with the iron.



Orientation

U1, U2 and U3: pin 1 is upper left.
D2: cathode band goes left.

C1/C2 are the purchased **CL31B106KLHNNNE** capacitors.

Leave these flat

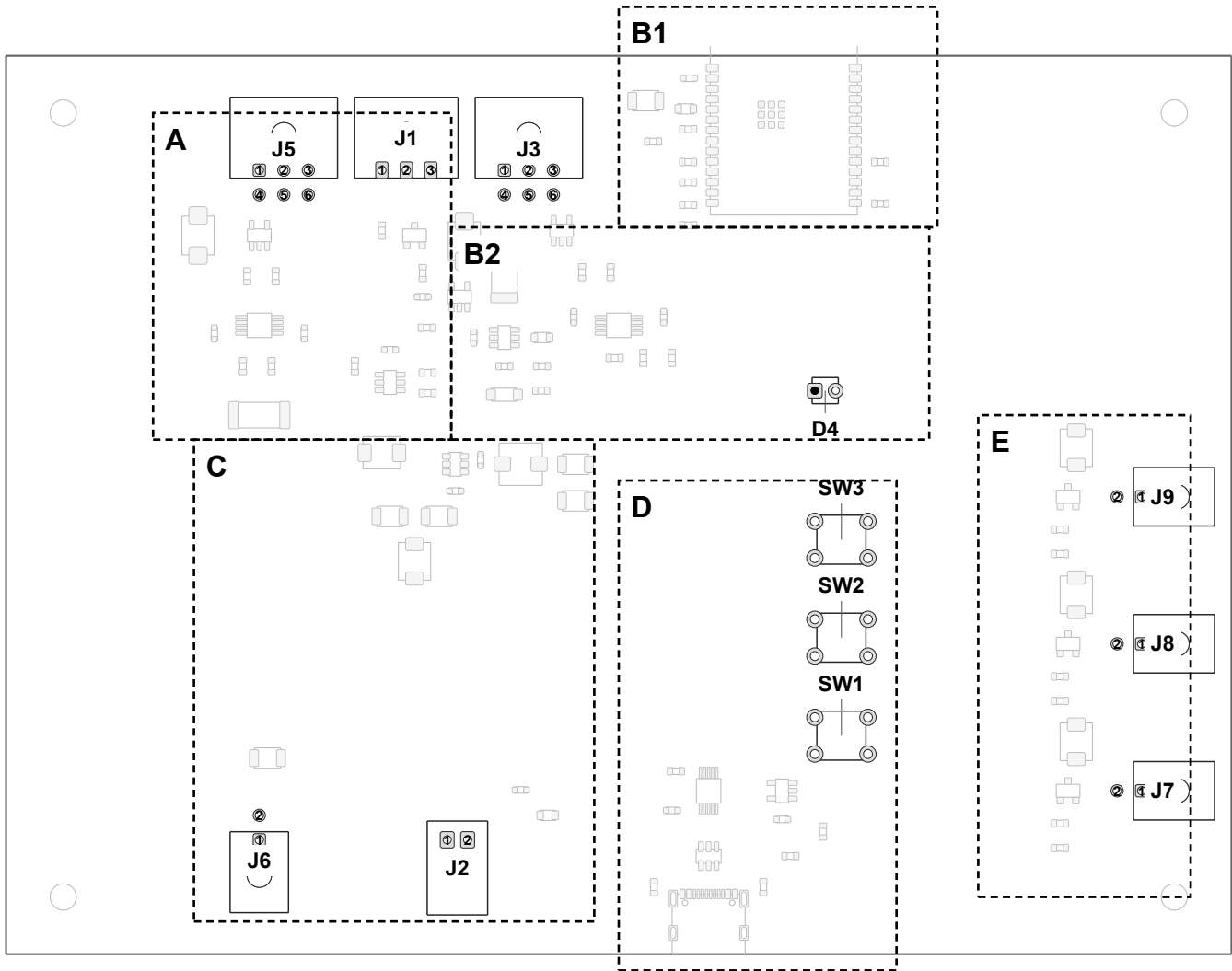
J1 is six wire lands, not a header. E1 is built-in sensing copper on the back. Put no paste, tin or components on the sensing face.

ENLARGED ELECTRONICS AREA / TOP OF THE 18 × 64 mm BOARD

PLACE AT	EXACT BAG / PART NUMBER	VALUE / BODY	QTY
C1, C2	CL31B106KLHNNNE	10uF · 1206 / 3216	2 SMT
C3	C1608X7R1H104K080AA	100nF · 0603 / 1608	1 SMT
C4	C2012X7R1E105K125AB	1uF · 0805 / 2012	1 SMT
D2	STPS2L40U	SMB / DO-214AA	1 SMT
R1, R2	ERJ-3EKF22R0V	22Ω · 0603 / 1608	2 SMT
R3, R4	ERJ-3EKF2701V	2.7kΩ · 0603 / 1608	2 SMT
R5	ERJ-3EKF1002V	10kΩ · 0603 / 1608	1 SMT
R6	ERJ-3EKF3011V	3.01kΩ · 0603 / 1608	1 SMT
U1	FDC1004DGSR	VSSOP-10 (DGS)	1 SMT
U2	TPS7A2433DBVR	SOT-23-5 (DBV)	1 SMT
U3	ESDS312DBVR	SOT-23-5 (DBV)	1 SMT

Controller / iron parts + area map

White board. ESP antenna at the top; USB at the bottom. Reflow first; fit the 12 iron-only parts below afterward.



A: relay + reservoir · B1: ESP · B2: tank interface · C: power · D: USB · E: pumps

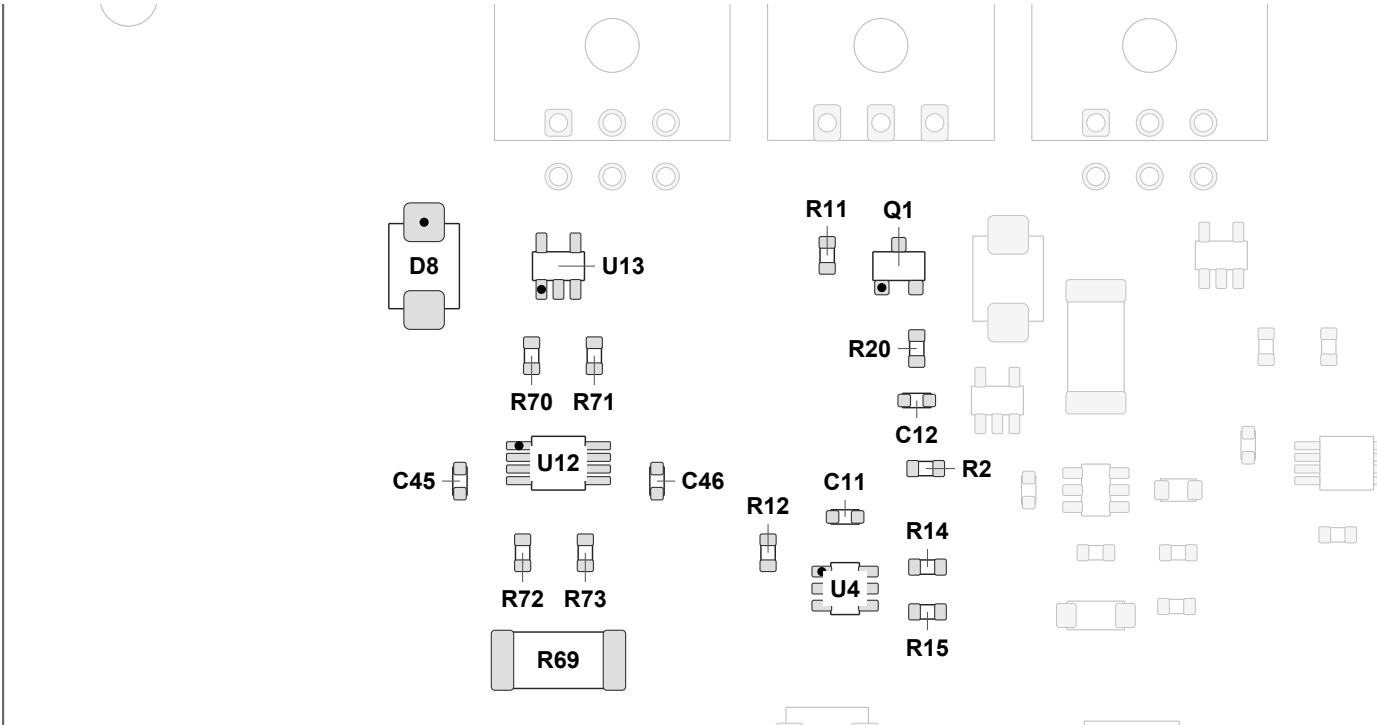
PLACE AT	EXACT BAG / PART NUMBER	VALUE / BODY	QTY
D4	HLMP-1790	3 mm through-hole LED	1 IRON
J1	0436500300	Micro-Fit 3.0, 3 positions, right angle	1 IRON
J2	S2B-XH-A	JST XH, 2 positions, side entry	1 IRON
J3, J5	0430450600	Micro-Fit 3.0, 6 positions, right angle	2 IRON
J6, J7, J8, J9	0430450200	Micro-Fit 3.0, 2 positions, right angle	4 IRON
SW1, SW2, SW3	B3F-1002-G	6 mm through-hole tactile switch	3 IRON

Iron-part orientation

J1/J3/J5 mouths up; J2/J6 down; J7–J9 right. D4 cathode left. Buttons top to bottom: SW3 RESET, SW2 BOOT, SW1 MAINTENANCE. Fit all 12 parts; TP1–TP14 and H1–H4 get no parts.

Controller A / Relay + reservoir interface

Stencil + hot bed. Iron-only parts are listed on page 4. Qty applies only to this map.



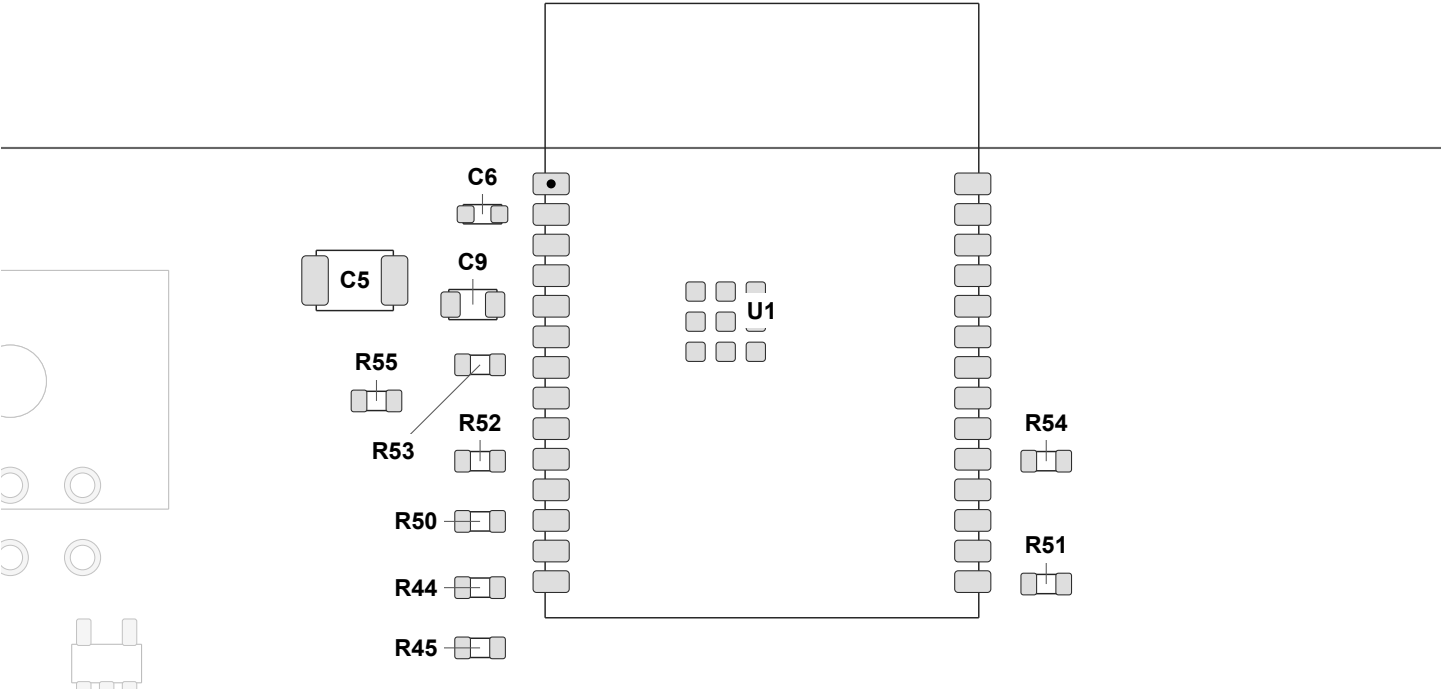
U4 and U12: pin 1 upper left. U13: pin 1 lower left. D8 band up. Q1 two-leg side down.

PLACE AT	EXACT BAG / PART NUMBER	VALUE / BODY	QTY
C11, C12, C45, C46	C1608X7R1H104K080AA	100nF · 0603 / 1608	4 SMT
D8	SMBJ7.0A	SMB / DO-214AA	1 SMT
Q1	AO3400A	SOT-23, 3 leads	1 SMT
R2, R11, R12	ERJ-3EKF1002V	10kΩ · 0603 / 1608	3 SMT
R14	ERA-3AEB9532V	95.3kΩ · 0603 / 1608	1 SMT
R15	ERA-3AEB103V	10kΩ · 0603 / 1608	1 SMT
R20	ERJ-3EKF1000V	100Ω · 0603 / 1608	1 SMT
R70, R71	ERJ-3EKF22R0V	22Ω · 0603 / 1608	2 SMT
R72, R73	ERJ-3EKF2701V	2.7kΩ · 0603 / 1608	2 SMT
R69	CRCW25126R80FKEGHP	6.8Ω · 2512	1 SMT
U13	ESDS312DBVR	SOT-23-5 (DBV)	1 SMT
U12	TCA9517ADGKR	VSSOP-8 (DGK)	1 SMT
U4	TPS3808G01DBVR	SOT-23-6 (DBV)	1 SMT

R14 is the precision 95.3 kΩ part; R15 is the precision 10 kΩ ERA part. Keep it separate from ordinary ERJ 10 kΩ resistors.

Controller B1 / ESP module + nearby passives

Stencil + hot bed. Iron-only parts are listed on page 4. Qty applies only to this map.



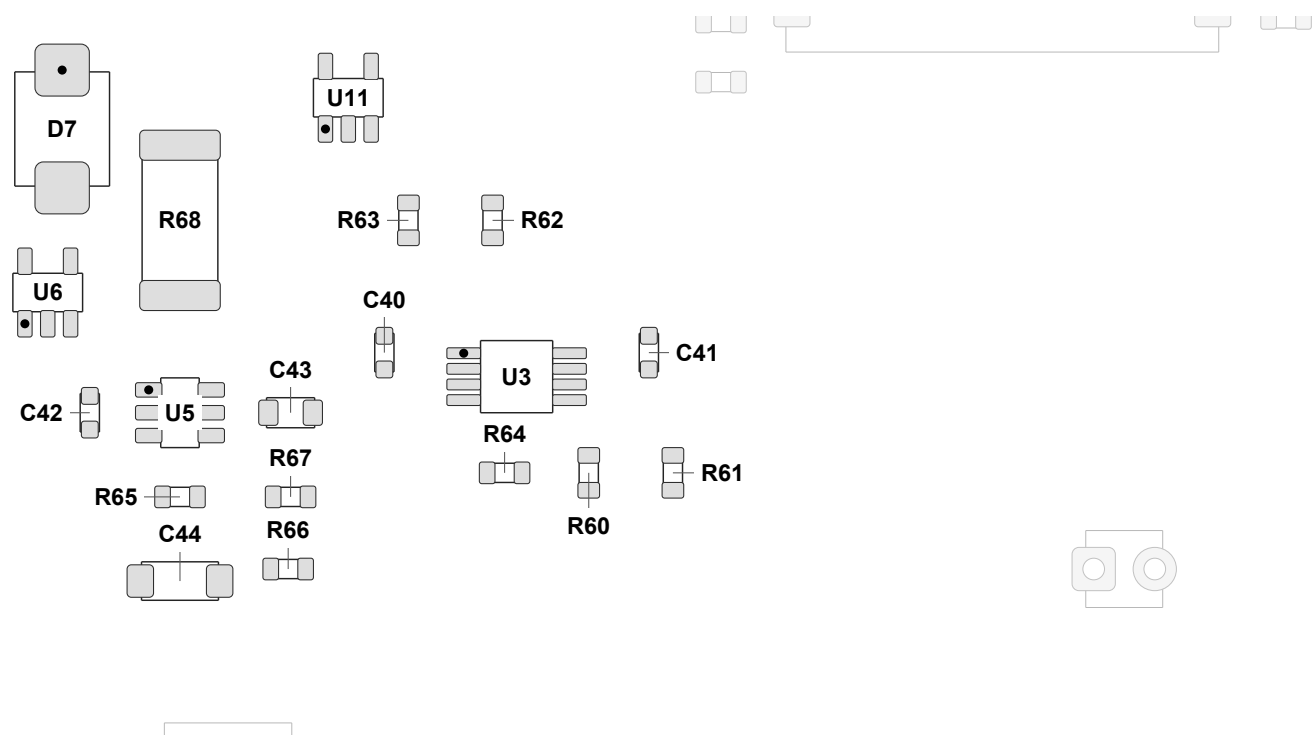
U1 pin 1 is the uppermost left perimeter pad. The antenna projects beyond the board's top edge.

PLACE AT	EXACT BAG / PART NUMBER	VALUE / BODY	QTY
C5	CGA6P3X7R1E226M250AB	22uF · 1210 / 3225	1 SMT
C6	C1608X7R1H104K080AA	100nF · 0603 / 1608	1 SMT
C9	C2012X7R1E105K125AB	1uF · 0805 / 2012	1 SMT
R50, R51, R52, R53	ERJ-3EKF1002V	10kΩ · 0603 / 1608	4 SMT
R44, R45	ERJ-3EKF22R0V	22Ω · 0603 / 1608	2 SMT
R54	ERJ-3EKF6800V	680Ω · 0603 / 1608	1 SMT
R55	CRCW0603330RFKEA	330Ω · 0603	1 SMT
U1	ESP32-C6-WROOM-1-N8	Castellated RF module	1 SMT

Print the existing nine underside ground openings. Reflow U1 with the other SMT parts; perimeter-only iron soldering would leave hidden grounds unsoldered.

Controller B2 / Tank sensor interface + LED

Stencil + hot bed. Iron-only parts are listed on page 4. Qty applies only to this map.



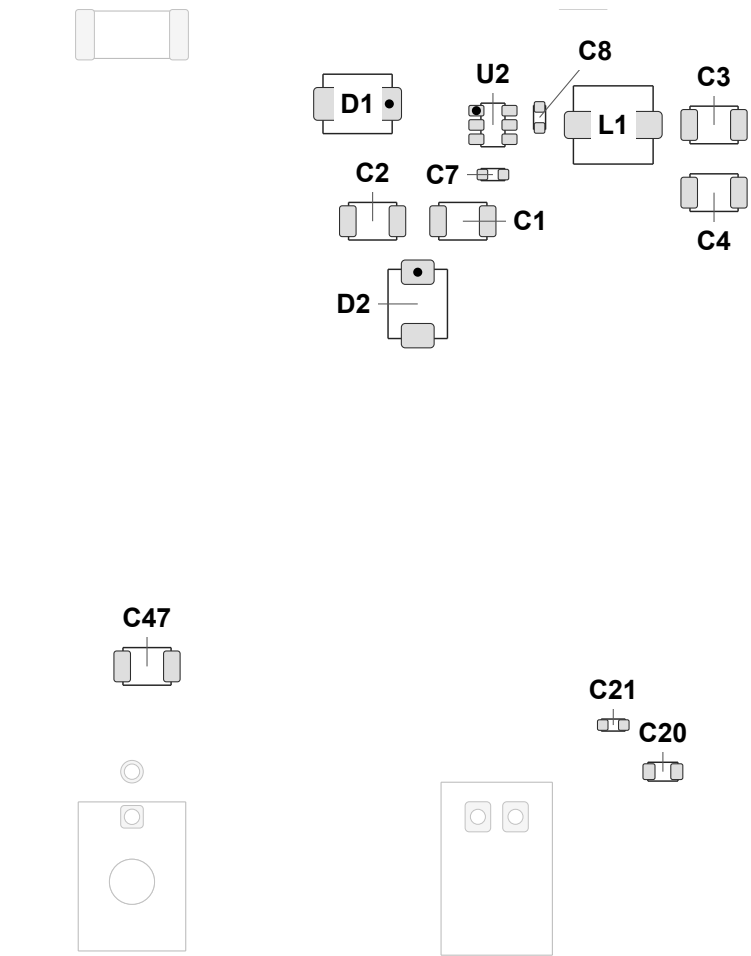
U3/U5 pin 1 upper left; U6/U11 pin 1 lower left. D7 band up. D4 cathode is the left lead.

PLACE AT	EXACT BAG / PART NUMBER	VALUE / BODY	QTY
C40, C41, C42	C1608X7R1H104K080AA	100nF · 0603 / 1608	3 SMT
C43	C2012X7R1E105K125AB	1uF · 0805 / 2012	1 SMT
C44	CL31B106KLHNNNE	10uF · 1206 / 3216	1 SMT
D7	SMBJ7.0A	SMB / DO-214AA	1 SMT
R64, R65, R66, R67	ERJ-3EKF1002V	10kΩ · 0603 / 1608	4 SMT
R62, R63	ERJ-3EKF22R0V	22Ω · 0603 / 1608	2 SMT
R60, R61	ERJ-3EKF2701V	2.7kΩ · 0603 / 1608	2 SMT
R68	CRCW25126R80FKEGHP	6.8Ω · 2512	1 SMT
U11	ESDS312DBVR	SOT-23-5 (DBV)	1 SMT
U3	TCA9517ADGKR	VSSOP-8 (DGK)	1 SMT
U5	TPS2553DBVR	SOT-23-6 (DBV)	1 SMT
U6	SN74LVC1G08DBVR	SOT-23-5 (DBV)	1 SMT

D4 is the purchased HLMP-1790 LED. Fit it with the iron after reflow. J3's mating mouth faces the top edge.

Controller C / Power + service connectors

Stencil + hot bed. Iron-only parts are listed on page 4. Qty applies only to this map.



Orientation

U2 pin 1 upper left. D1 band right; D2 band up. J2 and J6 mating mouths face the bottom edge.

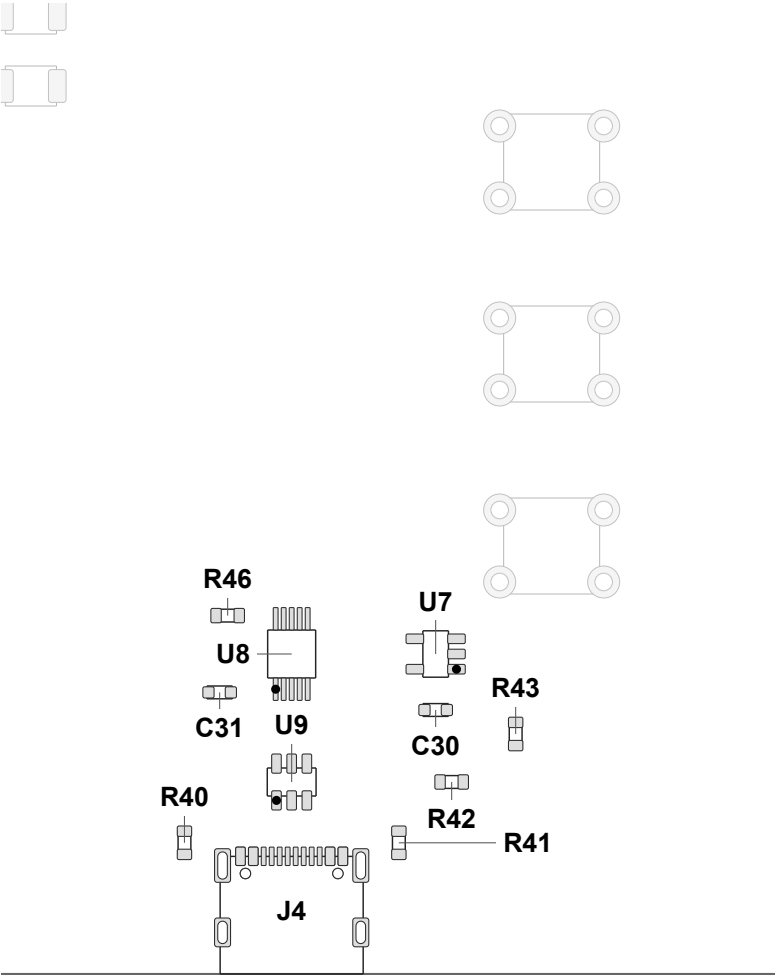
Assembly

Controller U2 is AP63203WU-7, the 3.3 V regulator. Mains U2 is AP63205WU-7. Their packages look the same; keep their bags separate.

PLACE AT	EXACT BAG / PART NUMBER	VALUE / BODY	QTY
C1, C2, C3, C4, C47	CGA6P3X7R1E226M250AB	22uF · 1210 / 3225	5 SMT
C7, C8, C21	C1608X7R1H104K080AA	100nF · 0603 / 1608	3 SMT
C20	C2012X7R1E105K125AB	1uF · 0805 / 2012	1 SMT
D1, D2	STPS2L40U	SMB / DO-214AA	2 SMT
L1	SRP5030TA-4R7M	4.7µH · SRP5030TA	1 SMT
U2	AP63203WU-7	TSOT26, 6 leads	1 SMT

Controller D / USB + pushbuttons

Stencil + hot bed. Iron-only parts are listed on page 4. Qty applies only to this map.



Orientation

U7 pin 1 lower right; U8/U9 pin 1 lower left. USB mouth faces the bottom edge.

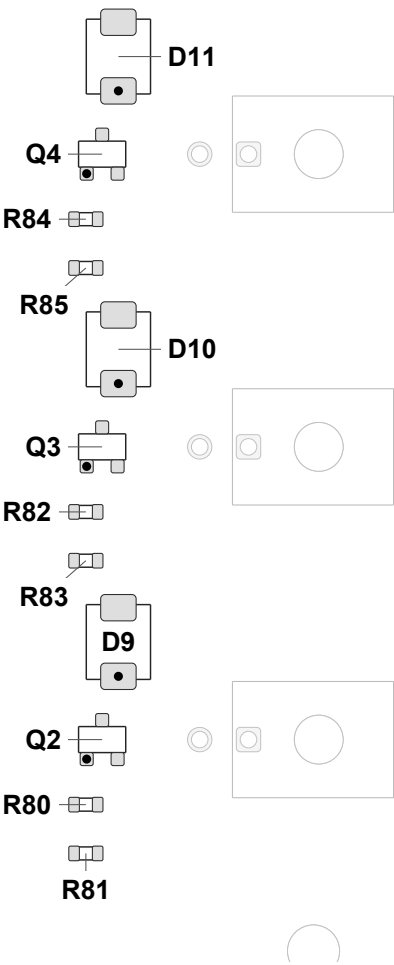
Assembly

J4: reflow the signal contacts, then iron-solder all four shell tabs. Keep the supplied stencil's duplicate-contact exclusions. Buttons top to bottom: SW3 RESET, SW2 BOOT, SW1 MAINTENANCE.

PLACE AT	EXACT BAG / PART NUMBER	VALUE / BODY	QTY
C30, C31	C1608X7R1H104K080AA	100nF · 0603 / 1608	2 SMT
J4	USB4105-GF-A	USB-C, 16 contacts, top mount	1 SMT*
R46	ERJ-3EKF1002V	10kΩ · 0603 / 1608	1 SMT
R40, R41	ERJ-3EKF5101V	5.1kΩ · 0603 / 1608	2 SMT
R42	ERJ-3EKF1001V	1kΩ · 0603 / 1608	1 SMT
R43	ERJ-3EKF1003V	100kΩ · 0603 / 1608	1 SMT
U7	SN74LVC1G14DBVR	SOT-23-5 (DBV)	1 SMT
U8	FSUSB42MUX	MSOP-10	1 SMT
U9	USBLC6-2SC6	SOT-23-6	1 SMT

Controller E / Three pump drivers

Stencil + hot bed. Iron-only parts are listed on page 4. Qty applies only to this map.



Orientation

D9/D10/D11 cathode bands point down. Q2/Q3/Q4 two-leg sides point down; lone drain legs point up.

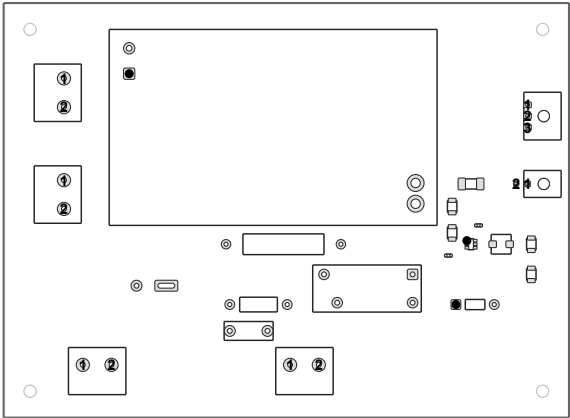
Assembly

Populate all three driver circuits. Iron-solder J7–J9 with mouths toward the right edge. These accessory outputs remain unplugged in the present skimmer build.

PLACE AT	EXACT BAG / PART NUMBER	VALUE / BODY	QTY
D9, D10, D11	STPS2L40U	SMB / DO-214AA	3 SMT
Q2, Q3, Q4	AO3400A	SOT-23, 3 leads	3 SMT
R81, R83, R85	ERJ-3EKF1002V	10kΩ · 0603 / 1608	3 SMT
R80, R82, R84	ERJ-3EKF1000V	100Ω · 0603 / 1608	3 SMT

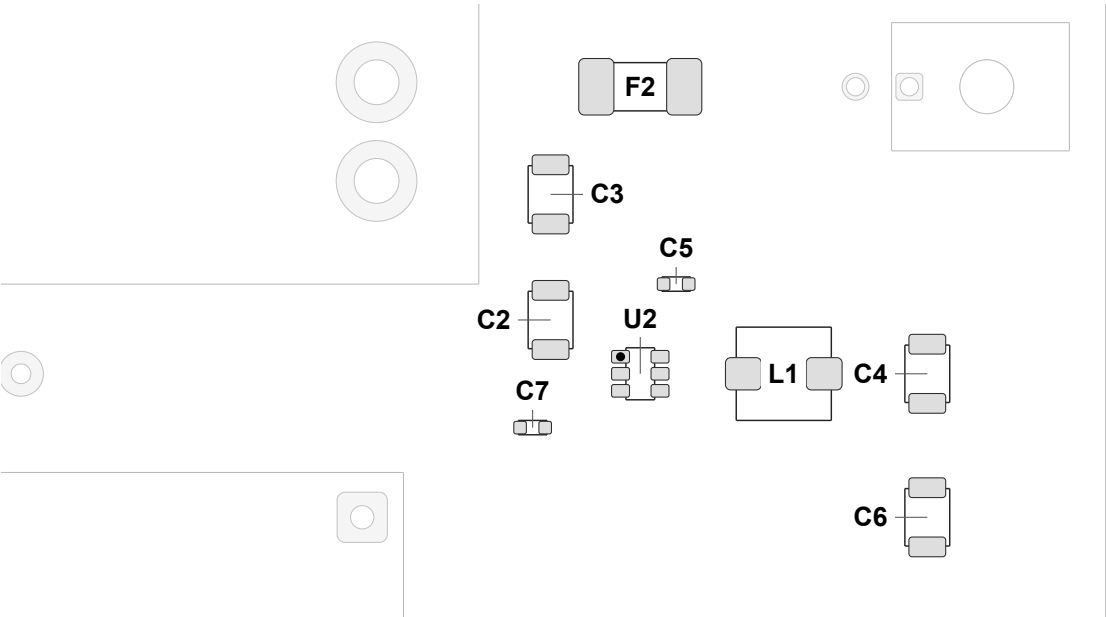
Mains / nine SMT parts first

Blue board. Reflow the small power-supply area before fitting the tall components.



Find this corner

The SMT group is on the right, above the relay. U2 is **AP63205WU-7 (5 V)**. Pin 1 goes upper left. F2 is the flat 3 A SMT fuse.



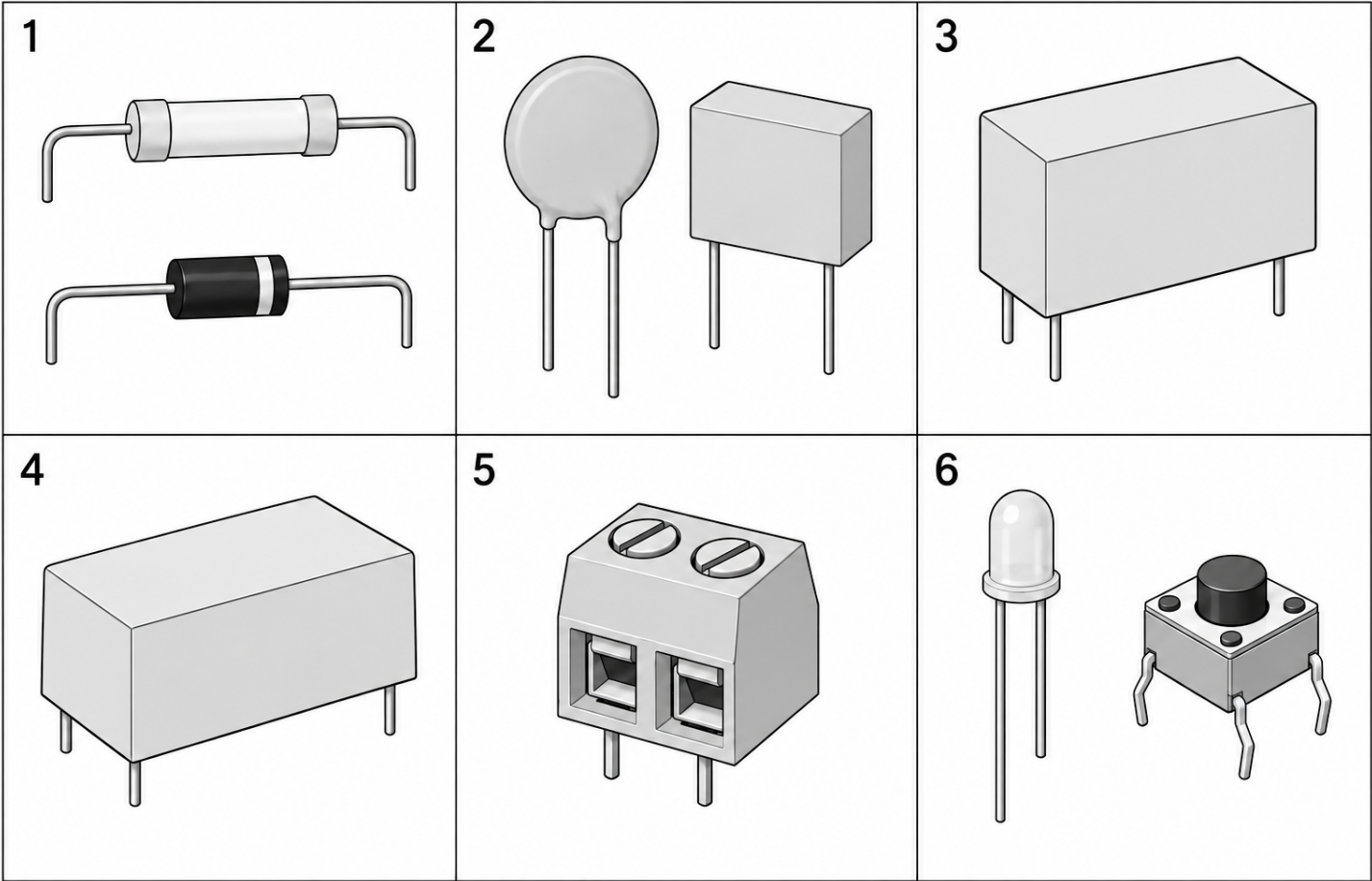
PLACE AT	EXACT BAG / PART NUMBER	VALUE / BODY	QTY
C2, C3, C4, C6	CGA6P3X7R1E226M250AB	22uF · 1210 / 3225	4 SMT
C5, C7	C1608X7R1H104K080AA	100nF · 0603 / 1608	2 SMT
F2	0451003.MRL	NANO ² 451 SMT fuse	1 SMT
L1	SRP5030TA-4R7M	4.7μH · SRP5030TA	1 SMT
U2	AP63205WU-7	TSOT26, 6 leads	1 SMT

Keep the hot-bed pass flat

Leave U1, K1, RV1, F3, C1, D1, R1 and all six headers off until reflow is finished and the board has cooled.

Recognize the iron-soldered parts

Generic body shapes, not pin-layout drawings. Exact positions and orientations are on the board maps.



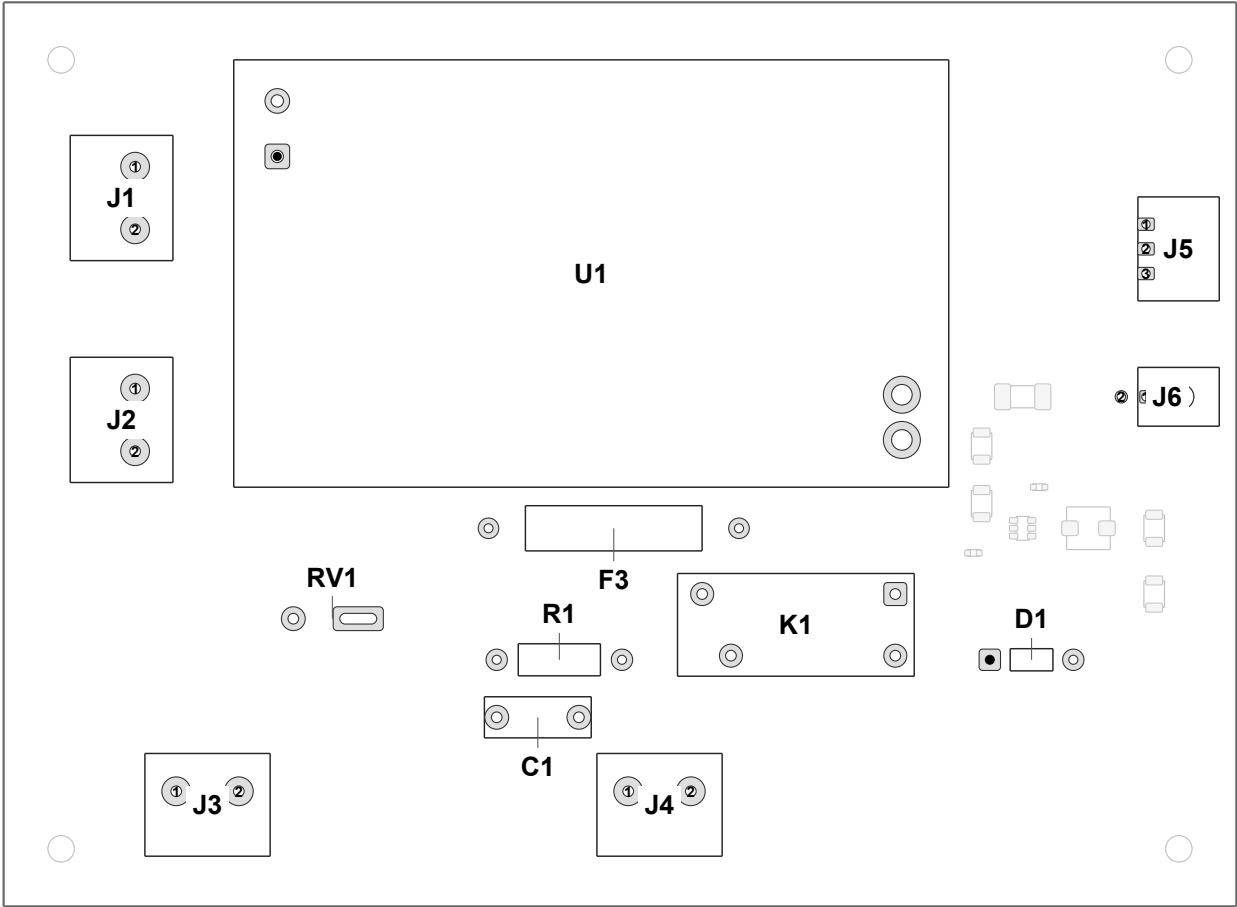
PICTURE	EXACT PART	PLACE AT
1 Axial bodies	PR02FS0201000KR500 · 100 Ω fusible resistor 1N4007-E3/54 · banded diode	Mains R1 Mains D1
2 Disc + box	TMOV14RP175E · thermal MOV B32921C3473K000 · 47 nF safety capacitor	Mains RV1 Mains C1
3 Relay	G5RL-1A-TV8 DC5 · 5 V coil	Mains K1
4 AC/DC module	IRM-45-12 · large encapsulated PSU	Mains U1
5 Screw terminals	1868076 · two positions, right-angle wire entry	Mains J1–J4
6 LED + buttons	HLMP-1790 · green 3 mm LED B3F-1002-G · four-lead pushbuttons	Controller D4 Controller SW1–SW3

Other iron parts

F3 is the leaded 0215001.MXEP fuse. Molex and JST board connectors are keyed plastic housings with solder tails. Use each map's mating-direction cue and the board's body outline; do not force a rotated connector.

Mains / fit the tall parts

Iron only. All parts sit on the component side; solder their leads from the back.

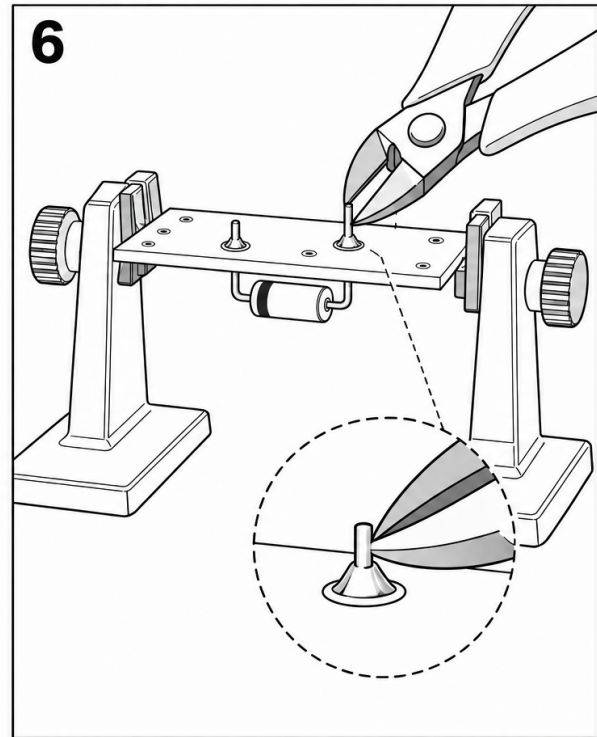
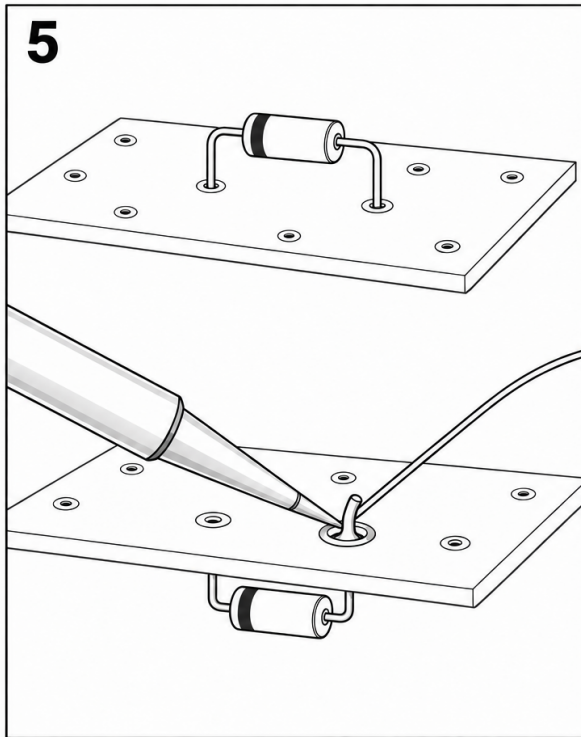


PLACE AT	EXACT BAG / PART NUMBER	VALUE / BODY	QTY
C1	B32921C3473K000	47nF · X2 radial film capacitor	1 IRON
D1	1N4007-E3/54	DO-41 / DO-204AL axial	1 IRON
F3	0215001.MXEP	215 axial leaded fuse	1 IRON
J1, J2, J3, J4	1868076	MKDS 5/2-7,62, 2-position fixed terminal	4 IRON
J5	0436500300	Micro-Fit 3.0, 3 positions, right angle	1 IRON
J6	0430450200	Micro-Fit 3.0, 2 positions, right angle	1 IRON
K1	G5RL-1A-TV8 DC5	Through-hole relay, 4 pins	1 IRON
R1	PR02FS0201000KR500	100Ω · PR02FS axial, 2 W	1 IRON
RV1	TMOV14RP175E	Bulk two-lead thermally protected MOV	1 IRON
U1	IRM-45-12	Through-hole encapsulated PSU	1 IRON

D1 band left. Relay coil pins right. PSU AC pins left, DC pins lower right. J1/J2 mouths left; J3/J4 mouths down; J5/J6 mouths right. See the next page for F3 and RV1 seating.

Iron work / fit, solder, trim

After each board has cooled, fit low through-hole parts first and the largest bodies last.



5 FIT + SOLDER

Fit one part, secure it, and turn the board over. Heat pad and lead together, then feed solder into the joint. Keep the board supported clear of its components.

6 TRIM + FINISH

Cut the lead above the solder fillet; hold the offcut. Keep mains lead/solder projections within 2 mm of the underside. Clean SMT work before installing the non-washable switches and relay.

F3: axial T1A fuse / 0215001.MXEP

Seat the fuse at least 1.5 mm above the board. Make bends more than 1 mm from its end caps. Its maker specifies an iron at 350 °C ±5 °C for at most 5 seconds. Do not reflow it.

RV1: thermal MOV / TMOV14RP175E

Fit after reflow. Keep its unbent, staggered leads; rotate the upright body to match the round hole and slot. Support 0.5–1.0 mm above the board and heat-sink each lead while soldering. Do not force or reshape the leads.

RV1 final fit: upright within 5°, yaw at most 35°, top at most 26 mm above the PCB; entire body/leads/solder stay inside its 27 × 27 mm assembly envelope. Keep solder and offcuts out of isolation gaps.

Wire the three connections

One sensor lead and two internal harnesses. Circuit maps below are symbolic, not housing-face views.

INTERNAL BOARD-TO-BOARD HARNESSES



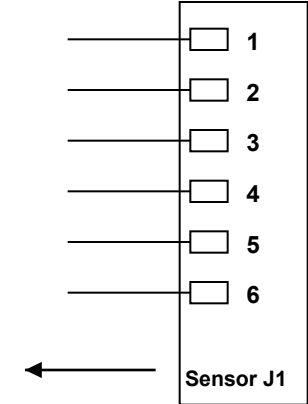
HARNESS	SAME CIRCUIT AT BOTH ENDS	WIRE	EXACT BAG / FITTED SET
Mains J5 to controller J1	1 → 1: +5 V 2 → 2: GND 3 → 3: COIL_DRAIN	22 AWG Red / black / blue	2 × 0436450300 housings 6 × 0430300007 contacts (Molex 43645-0300 / 43030-0007)
Mains J6 to controller J6	1 → 1: +12 V 2 → 2: GND	20 AWG Red / black	2 × 0430250200 housings 4 × 0430300007 contacts (Molex 43025-0200 / 43030-0007)

Keep the 12 V pair together through its own harness. J7–J9 on the controller are future pump outputs; their pin 2 is a switched return. Feed the controller at **J6**.

SENSOR J1 → CONTROLLER J3

Fit **1 × 0430250600** housing and **6 × 0430300007** contacts at the controller end (Molex 43025-0600 / 43030-0007). The sensor end solders directly to J1.

OUTWARD FACE



Exit left

Land 1 is at the top.
Solder only on this face.

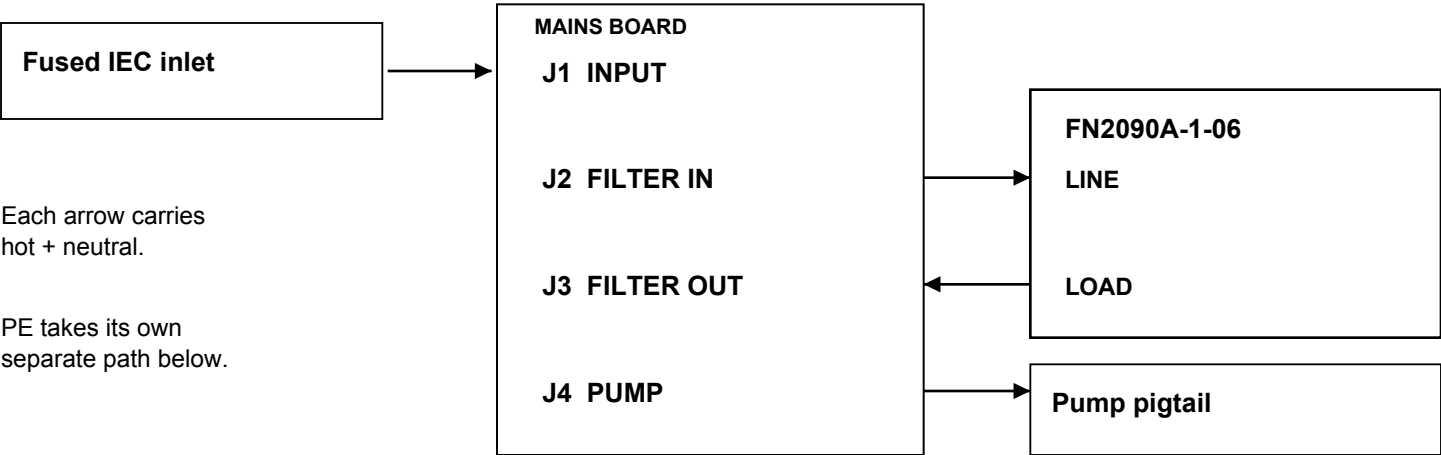
SENSOR LAND → J3	FUNCTION	TWISTED PAIR
1 → 1	+5 V sensor supply	1 + 4
2 → 2	SDA	2 + 5
3 → 3	SCL	3 + 6
4 → 4	Power ground return	1 + 4
5 → 5	SDA ground return	2 + 5
6 → 6	SCL ground return	3 + 6

Use **24 AWG, three twisted pairs**. Identify the actual cable pairs and label both ends 1–6 before separating them. Keep all three ground returns. Cut back and insulate any overall foil or drain at both ends.

203.2 mm maximum, including the ends
Measure from sensor solder termination to controller crimp, including internal routing and strain relief. Cut to the installed route. The ordered cable's model and specifications are unrecorded; identify its pairs and confirm contact fit before termination. Match circuit numbers, not an assumed left-to-right cavity order.

Connect mains, filter and PE

Keep every power source disconnected. This is a connection map, not a literal wire route.



MAINS BLOCK	PIN 1 · HOT (L)	PIN 2 · NEUTRAL (N)
J1 · INPUT	From inlet fused L output	From inlet N
J2 · FILTER IN	To filter LINE L	To filter LINE N
J3 · FILTER OUT	From filter LOAD L	From filter LOAD N
J4 · PUMP	To output-pigtail hot	To output-pigtail neutral

PE bypasses the boards. Three separate branches meet at one covered M4 metal junction. Keep PE separate from neutral; neither PE nor neutral is switched.

PE BRANCH	EQUIPMENT END	M4 JUNCTION END
Inlet PE	TE 2-520184-2 FASTON	TE 31886 ring
Filter PE blade	TE 2-520184-2 FASTON	TE 31886 ring
Output pigtail PE	Existing pigtail conductor	TE 320551 ring

Offboard parts: Schurter **6200.23** inlet; Littelfuse **0215002.MXP** 2 A fuse; **3-125-661 (RC320)** rear cover; **FN2090A-1-06** filter; **1112.048.005350** output pigtail; **M3231** gland and **8463** locknut.

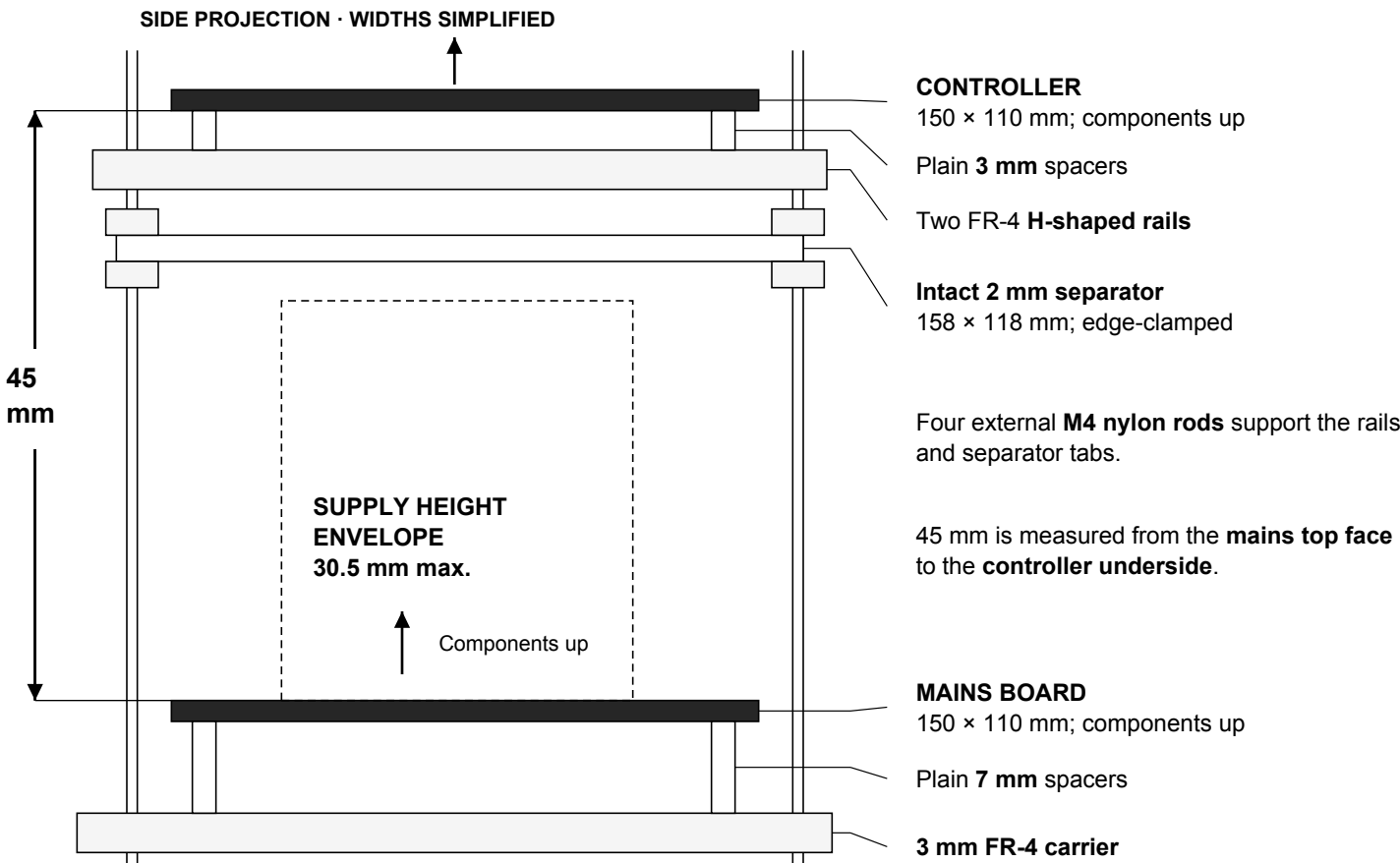
Fit **8 × TE 2-520184-2** insulated 6.35 mm receptacles: inlet L/N/PE (3), filter LINE/LOAD L/N (4), filter PE (1). Use **18 AWG, 600 V, 105 °C** mains/PE wire. The four **Phoenix Contact 1868076** PCB blocks take wire directly.

Finish the lower wiring before fitting the separator

Strip **8 mm**; tighten to **0.5–0.6 N·m**, supporting the terminal body. Keep filter LINE/LOAD apart and **8 mm** between primary wiring and SELV. Clamp PE rings metal-to-metal on separate insulating support; exclude plastic from the electrical clamp. Fit both covers.

Combine the finished boards

Controller above mains, both component sides up. Fit all primary wiring before closing the stack.



COMBINE IN THIS ORDER

STEP	ACTION
1	Finish mains J1–J4 wiring and all three PE branches. Fit the inlet and PE covers while the terminal screws remain accessible.
2	Mate both internal harnesses at mains J5 and J6. Carry them around the isolated edge, clear of primary wiring and the antenna reserve.
3	Capture the intact separator between its corner tabs. Fit the H rails, plain 3 mm spacers and controller using insulating through-bolts.
4	Mate controller J1 (5 V/coil), J6 (12 V) and J3 (sensor). Leave future reservoir J5 and pump outputs J7–J9 empty.

Use the four aligned M3 PCB positions. Keep the separator intact, with no holes or antenna notch. Use single through-bolts through plain spacers; include washers in the effective spacing. The 45 mm face gap is **not** a 45 mm spacer length.